

synthesisers, movement-to-audio converters, etc, so people can create music without depending on sight, speech or even hearing!

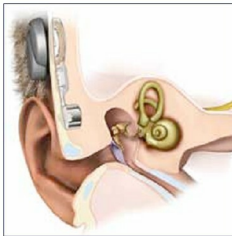
Then, there are people such as Doug Briggs, the pioneer of the term Assistive Music Technology (AMT), who are constantly developing new products or customising existing ones to enable special people to tap the talent within them or to help established musicians who are suddenly crippled by an illness or accident.

Briggs, along with his Adaptive Music Technology Research Group (AMTRG) at Huddersfield University, has adapted many pieces of existing technology for special needs education. A graduate student at their centre, for example, has developed an iPad app with a very simple and intuitive interface for controlling surround sound by drawing a path, then manipulating it with the iPad's two-finger squeeze or three-finger rotation. It is easy for everybody to use, and can also help a person who can move only three fingers to control the mix.

In a recent article in *Sound on Sound* (www.soundonsound.com), Doug Briggs explains that AMT involves three stages: input stage, processing stage and output stage. This is kind of similar to a typical modern digital recording situation where the input corresponds to playing an instrument, processing to sequencing and/or processing, and the output to listening to your mix.

A hearing aid that relies on vibrations

Med-El recently launched Bonebridge—a vibration-based hearing implant developed in collaboration with a London-based hospital. Bonebridge is surgically embedded under the skin, into the



Bonebridge—a hearing implant launched by Med-El

temporal bone. An external microphone worn under the hair captures sound and sends it to the implant, which transforms the noise into vibrations. The vibrations are transmitted to the inner ear, where they are processed into natural sound.

With Amadé BB audio processor, Bonebridge features the latest signal processing technology for improved auditory experience. As the audio processor can be renewed, the Bonebridge system can also be upgraded with state-of-the-art technologies, even many years after the implant surgery.

The specialty of this device is that it enables sound transmission directly to the inner ear by means of bone conduction. It is a suitable solution for people in whom sound cannot be transferred to the inner ear via the natural path of hearing through the outer and the middle ear. Plus, since the microphone is worn under the hair, it is not seen outside either.

The device has been approved for medical use in Canada, and costs between \$3000 and \$5000.

An all-new mobility platform

Matia Robotics has developed the Tek Robotic Mobilisation Device (Tek RMD) as a brand new mobil-

ity platform that brings exciting new possibilities for individuals with paraplegia and other walking disabilities to move around. It enables them to independently and safely sit, stand and navigate environments that were once inaccessible. It also overcomes many of the disadvantages of wheelchairs.

For example, wheelchairs can only be front-mounted. This is a problem because when a person is sitting, his front side is always free and his body is always resting on his back. So he has to lift his body with his arms and throw himself onto the wheelchair in order to transfer himself from a bed or a chair. This motion of throwing the body is difficult, dangerous and almost impossible to do without assistance, especially for older or overweight people.

On the other hand, Tek RMD can easily be summoned to come near the person using a remote. Then the user can easily pull the device closer, and mount it directly from its back.

It also enables people to sit or stand without much effort. So paraplegics can even use it to assist in common physiotherapy. The suspension system containing a gas spring balances the weight of the user so that standing up requires just a gentle pull.

From the moment they wake up, users can board the device and stand up without needing assistance from others. This comfort encourages the users to stand up countless times, and do various daily activities while standing up. It also includes braces to ensure a proper standing position and an advanced balancing system to avoid falls.

Presently, Tek RMD is also the world's smallest motorised standing movement device. It is only 39 cm wide and 75 cm long, so users can pass through narrow spaces too.

It is truly amazing to see how technology is helping make the world truly accessible to all. If you have developed, use or know of such technologies, do write to us or e-mail us to let more people know of the innovation. ●



Tek Robotic mobilisation device (Courtesy: www.matiarobotics.com)

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